

PROTOCOL: PCR Pooling

v 1.0

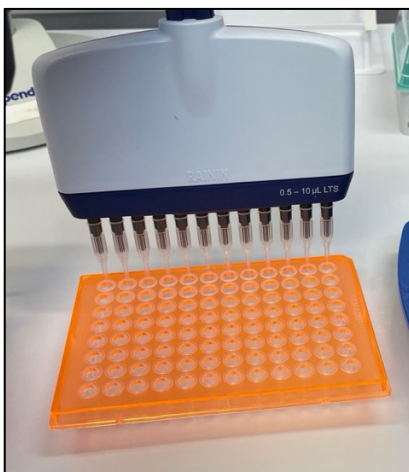
This protocol provides technical instructions for combining all PCR reactions into a single tube.

🧬 Bullet 🧬 points show tips for executing the protocol.

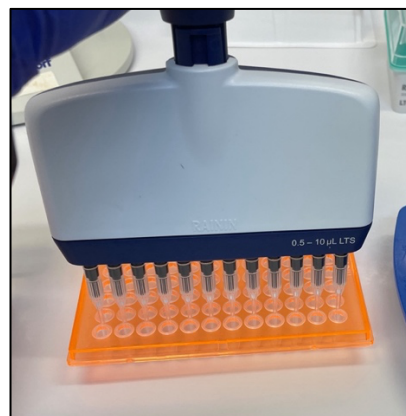
(1) Plate Pools: Pool each PCR plate into its own 1.5 mL tube

🧬 *Although this protocol can be completed entirely with a single-channel pipette, we recommend using a 12-channel or 8-channel pipette where possible for efficiency.*

- a. Obtain all ONTOLOGY™ plates that will be multiplexed into this run. Each plate must have completed PCR and, ideally, had some or all wells checked on a gel.
- b. Using a multi-channel pipette, combine all rows into a single row (if 4 μL was taken to run on a gel previously, the remaining volume per well is **8.5 μL** ; otherwise, the full PCR volume of **12.5 μL** remains).



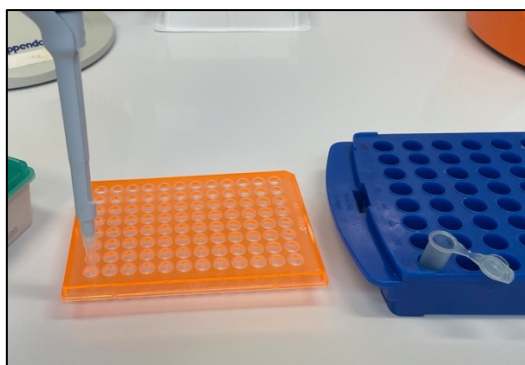
Use a multi-channel pipette to obtain all samples from row A



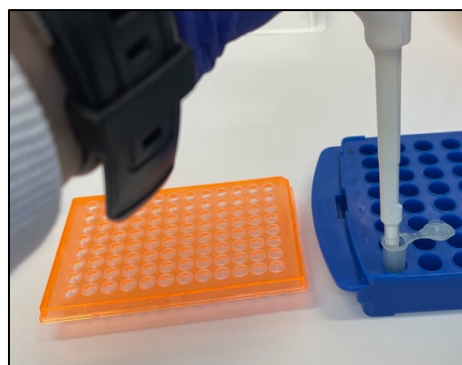
Dispense row A samples into row H; repeat for rows B–G



- c. Using a single-channel pipette and setting the pipette to collect the entire volume in each well, collect the combined row into a single 1.5 mL single tube.



Use a single-channel pipette to obtain pool in H1



Transfer H1 pool to destination 1.5 mL tube; repeat for H2–H12

- d. Mix well by pipetting after transfers are complete.
- e. Label tube carefully with plate ID and other relevant information
- f. Repeat a–d for all plates.

(2) Run-wide pool: Pool part of each plate pool into a tube for library preparation

§ The amount of each plate pool added to the Run-wide ‘pool of pools’ varies depending on the number of plates in the Run—consult the table below to determine the amount of each plate pool to add to the Run-wide pool. ONTOLOGY™ will also calculate the volume for you.

- a. Gather all plate pools
- b. Using a single-channel pipette, add a portion of each plate pool to a single tube that will become your Run-wide pool. Consult the table below and add the volume in (B) that corresponds to the number plates in your run (A).

(A) # plates in Run	(B) Volume per plate pool (μL)	Final volume (μL)
1	200	200
2	100	200
3	66.7	200
4	50	200
5	40	200
6	33.3	200
7	28.6	200
8	25	200
9	22.2	200
10	20	200

- c. Carry this Run-wide pool forward into library preparation.

